

Response to Reviewers' Comments, GMD D2017-243

We thank all three reviewers for their comments. We have tried address all of them, and our response is below. Reviewers' comments are italicized and response is in roman.

Response to Interactive Comment from Dann Mitchell

Many thanks for your remarks. Our response is as follows.

General Comments

1. *The paper would benefit from having an appendix of terms...or alternatively a brief short explanation of terms in the main text.*

We've gone with the option of explanations in the text. The correlated-k term was specifically mentioned, and here we have added: 'The 'correlated-k' method, with k being the absorption coefficient, is a means to efficiently calculate radiative transfer over a broad spectral range by collecting together wavenumber intervals with similar spectral properties and by supposing that these spectral properties are correlated from one level to another. A relatively small set of absorption coefficients can then be chosen to be representative of the absorption coefficients for all frequencies, leading to an enormous speed-up over line-by-line calculations and much better accuracy than traditional band methods that more simplistically just group together similar wavenumbers.' The references given in the text provide more detail.

2. *Python can be a nightmare for version control with syntax changing dramatically... Will the Python wrappers be frozen at a certain version? [Is] this different from point 4 on page 15?*

That is a fair point. Python did indeed change from Python 2 to Python 3 causing some disruption. There is an interesting blog about this by one of the Python core developers at:

<https://www.curiousinefficiency.org/posts/2014/08/python-4000.html>

Essentially, Python 4 will not change significantly from Python 3, and in any case Python 4 will not appear for quite some time — it now appears that the release after Python 3.9 will be Python 3.10. Our scripts currently support Python 2 and Python 3, but it may be that Python 2 support is dropped at some stage if there are advantages in having only Python 3, and there will be an official end-of-life of Python 2 in 2020. After that we don't expect there to be significant changes in Python syntax for the foreseeable future, but we would try to keep up to date. On that timescale changes in Fortran may also affect compilation of the core code. This issue is different from point 4; there we are remarking that the Python scripts have the ability to extract an old version of the model from the repo and compile and run it.

Minor Comments

1. *Affiliation* – fixed.
2. *Explanation of Isca* – added.
3. *Studying GFD of other planets can help with Earth* – Of course we agree, and think it is clear.
4. *Shortwave band treats all solar radiation.* – We are not sure what the problem is with this statement. We don't say it treats all radiation, just all *solar* radiation, and shortwave is a common term for solar radiation.
5. *A reasonable stratospheric polar vortex.* – We have removed 'reasonably realistic'.
6. *Importance of IR for planetary atmospheres.* – Yes, right. nearly all models, including idealized ones, have IR included, and here we are making the explicit point about the two-band scheme being an improvement over a grey model.
7. *Water ice vs CO₂ ice.* – Here we mean water ice and this is clarified. We haven't tried to model CO₂ ice, such as might be found on Mars, with the model.
8. *More modest changes.* – we rephrase to make this clearer.
9. *We have configured the thermal relaxation parameters.* – These are fundamental changes in the Fortran code. However, now that they are done, the user may change the parameters using the Python interface. Now clarified.

Response to Interactive Comment from Anonymous Referee #1

Many thanks for your remarks. Our response is as follows.

General Comments

Nothing is explicitly mentioned about topography.

Thanks for pointing that out. We have added a subsection to Section 5, (section 5.3) as the reviewer suggested.

I was less convinced at the simulations of Jupiter-like and Venus-like planets.

Fair enough, and there are two remarks we wish to make here.

- (i) Most importantly, in this paper we are just trying to demonstrate the capabilities of the framework, rather than discussing the science issues as to why, for example, the jets are where they are or presenting simulations that compare well with reality. That is an ongoing (and difficult!) task and we will be presenting such work in subsequent papers in the disciplinary literature (some are now in review or in preparation).
- (ii) Second, and more specifically, we have clarified our work in this paper (Section 8), and don't claim we are simulating Venus or Jupiter – we now talk about 'slowly rotating planets' and 'giant planets' for example. Indeed Titan is a better example for our slow rotation rate, although Venus is a better example for the increase in mass. Regarding the Venus-like case that only seemed to produce tropical super-rotation near the model lid: this may have been an unintended consequence of our graphics. We now show them using log-pressure co-ordinates (figure 9), as per a later comment by this reviewer. Regarding Jupiter, note that the Figure 8 showed vorticity, not velocity, and the jets are not especially weak. Other simulations we have done do give equatorial jets of a different nature and we will be presenting this work in future.

Specific Comments

- *There is more than one Geophysical Fluid Dynamics Laboratory.*
Quite so! The one founded by Raymond Hide is particularly revered by some of us, and there may be others. We've specified we mean the Princeton one.
- *P5. Reference to a paper in preparation. – Removed.*
- *P5 Eq (2) applies only in troposphere, and is presumably linearized.* Yes, it only applies in the troposphere. It can be extended upward isothermally, and we have

also clarified this. The Newtonian relaxation scheme itself is a linearization and we explicitly note that on P7.

- *P5 line 21 and line 22.* – Fixed.
- *P7, line 7. ‘The a’* – Fixed.
- *P9, Can RRTM not cope with more extreme conditions by extending the k-coefficients?*
Yes, it can in principle, as can any correlated-k scheme. However, we have not done this, and are not aware that others have either. We’ve clarified this at the end of section 4.3. Also, we have added a brief description of what a correlated-k scheme at the beginning of this section, in response to another reviewer’s comments.
- *P10, section 5. Needs a section on topography* Agreed, and we have added a subsection 5.3.
- *P 16, line 19., wording is rather clumsy.* Thanks for the close reading. We have rephrased along the lines you suggested.
- *P 17, Fig 9 and text. Vorticity not the easiest field to interpret.* We see your point, but please note our response to a related remark of yours in the general comments above.
- *P 17, figure 9 – jets in the Venusian case* We no longer call it the Venusian case and we now show a log pressure plot (figure 9) which clarifies these questions, and we state how many levels there are. Again note our response to your remark in the general comments above.
- *P18, line 2 parantheses wrong.* – Fixed.
- *P20, plans for a user forum?* That is a good idea and something that we have discussed, although at the moment we have no specific plans. If and when we have more external users we will certainly revisit that issue.
- *P22, a word missing to make sense.* – We slightly rephrased this sentence
- *P22, word ‘to’ missing.* – Fixed. Many thanks for noticing these things.

Response to Interactive Comment from Anonymous Referee #2

Many thanks for your remarks. Your main comment was that we had not properly given credit to past studies and you gave some examples. Of course that was not our intent. We might note that in the very first paragraph of the original paper we did write:

Various models at different levels of complexity have indeed been constructed (Frierson et al., 2006; Mitchell et al., 2006; O’Gorman and Schneider, 2008; Blackburn and Hoskins, 2013; Joshi et al., 2015, to name but a few).

To this list we have now added Fraedrich et al (2005, the very useful PUMA model). We have added a number of other references to papers that have a more planetary aspect, so that the introduction now contains:

A variety of models at different levels of complexity have in fact been constructed. Thus, to name but a few, Fraedrich et al. (2005a,b); Frierson et al. (2006); O’Gorman and Schneider (2008); Blackburn and Hoskins (2013) and Joshi et al. (2015) provide models of Earth’s atmosphere that are simplified in some way compared to a full GCM (of which there are a very great many). Similarly, regarding planetary atmospheres and again giving a limited sample, the Planet Simulator is a sibling of the PUMA model for planetary atmospheres (Fraedrich et al., 2005a); the SPARC model (Showman et al., 2009) uses the dynamical core of the MIT GCM but adds a more general radiation scheme appropriate for planetary atmospheres; the GFDL system has itself been used in a number of Earth and planetary settings (e.g., Mitchell et al., 2011; Schneider and Liu, 2009, and others); the UK Met Office Unified Model has been configured in various ways for both terrestrial exoplanets and hot Jupiters (Mayne et al., 2014; Boutle et al., 2017); the THOR model (Mendonca et al., 2016) solves the deep non-hydrostatic equations (as does the Unified Model) on an icosahedral grid and is designed to explore a range of planetary atmospheres; and CliMT (<https://github.com/CliMT/climt>) aims to provide a flexible Python based climate modelling toolkit. A number of quite comprehensive models, targeted at specific planets and similar in some ways to full GCMs of Earth, have also been developed.

We have mainly focussed on flexible models (rather than on papers that take a particular model and use it over a range of parameters) and we haven’t tried to reference comprehensive models or models of particular planets.